

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 is only adjacent to R1 via S0/0/0, which is a non-passive interface. R2 adjacent to R3 via S0/0/1 is still passive. R3 must go via R1(192.168.13.1) to reach the 192.168.2.0/24 network.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config-router)#router ospf 1

R2(config-router)#no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 from R3 direct link to
R2.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default is 100 Mbps, but modern interfaces like GigabitEthernet exceed 100 Mbps. Since OSPF cost must be an integer ≥ 1 , all links faster than 100 Mbps are assigned a cost of 1, making OSPF unable to distinguish them. Raising the reference bandwidth allows OSPF to assign different costs to GigabitEthernet vs FastEthernet links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For 192.168.3.0, $R1 \rightarrow R3$ via S0/0/1 (781) + R3's G0/0 (1) = **782**

For 192.168.23.0, $R1 \rightarrow R3$ via S0/0/1 (781) + R3's S0/0/1(64)
= **845**

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, because all serial interfaces now have bandwidth 128, so each serial link = 781. For 192.168.23.0, $R1 \rightarrow R3$ via S0/0/1 (781) + R3's S0/0/1(781) = 1562

Step 3: Change the route cost.

c.

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Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

the direct path R1 to R3 now costs 1565 (R1 s0/0/1) + 1 = 1566. The path via R2 costs 781 (R1 s0/0/0) + 781 (R2 s0/0/1) + 1 (R3's G0/0) = 1563. OSPF always picks the lowest-cost path, so traffic now flows through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

To ensure a unique, stable, and recognisable identifier for each router, preventing adjacency failures and routing instability caused by duplicate IDs.

2. Why is the DR/BDR election process not a concern in this lab?

OSPF only elects a DR/BDR on multi-access networks. However, all networks in the lab are point-to-point serial links. There is no multi-access network in which a single router must represent the entire network.

3. Why would you want to set an OSPF interface to passive?

To stop sending Hello packets on interfaces facing end devices (PCs) that can save traffic flow.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The aspect of the areas. OSPF can divide an AS into smaller, independent groupings called Areas, which all connect to the backbone Area 0. After that, routers only need to know the network map of their specific area rather than the entire system. It is flexible to scale up the network without recalculating the entire SPF.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

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EIGRP (AD = 90), since the router uses Administrative Distance (AD) to choose the protocol with the lowest AD when multiple routing protocols offer routes to the same network, .
